

Premature Graying and Associated Findings in Vitiligo

Premature graying has been reported in vitiligo patients and their close relatives. Auditory and visual disturbances may occur in some individuals. Rarely, aseptic meningitis can result from the destruction of leptomeningeal melanocytes.

Vitiligo-like depigmentation is occasionally observed in patients with malignant melanoma. This is thought to arise from a T-cell-mediated immune response against melanoma cells that cross-reacts with normal melanocytes. A similar depigmentation phenomenon can occur during autologous T-cell-based immunotherapy for melanoma and may be associated with a more favorable prognosis. Depigmented areas around primary tumors may resemble halo nevi, but widespread, distant vitiligo-like lesions can also develop.

Ocular Disease

Although most vitiligo patients do not report ophthalmologic symptoms, various ocular abnormalities may be present. Pigmentary changes in the iris and retina can occur. Choroidal abnormalities have been documented in up to 30% of patients, and iritis in approximately 5%. Uveitis is a common ocular manifestation. Exophthalmos may be seen in patients with coexisting Graves disease. Visual acuity is typically preserved.

Vogt-Koyanagi-Harada Syndrome

Vogt-Koyanagi-Harada (VKH) syndrome is a rare systemic autoimmune disorder mediated by T cells, characterized by vitiligo, uveitis, aseptic meningitis, dysacusis, tinnitus, poliosis, and alopecia. It is associated with other autoimmune conditions such as autoimmune polyglandular syndrome, hypothyroidism, Hashimoto thyroiditis, and diabetes mellitus.

VKH syndrome classically progresses through three phases:

1. **Meningoencephalic phase:** Preceded by fever, malaise, nausea, and vomiting, patients may develop headache, meningismus, seizures, muscle weakness, or paralysis.
2. **Acute ophthalmic phase:** Patients present with photophobia, eye pain, and changes in visual acuity. Uveitis, iridocyclitis, choroiditis, and retinal detachment may occur, potentially leading to complications such as cataracts and glaucoma.
3. **Chronic or convalescent phase:** Vitiligo, alopecia, and poliosis typically appear during this stage, although they may occasionally precede ocular or neurological symptoms.

Alezzandrini Syndrome

Alezzandrini syndrome is a rare condition characterized by unilateral facial vitiligo, poliosis, sensorineural deafness, and unilateral tapetoretinal degeneration. The etiology is not well understood, but autoimmune mechanisms—similar to those implicated in vitiligo and VKH syndrome—are suspected. Since its initial description in 1959 (in a patient with vitiligo, poliosis, and unilateral retinitis), only a few cases have been reported.

Laboratory Tests

The diagnosis of vitiligo is primarily clinical. However, due to its association with other autoimmune diseases, screening laboratory tests are often useful. Recommended tests include:

- Thyroid-stimulating hormone (TSH)
- Antinuclear antibody (ANA)
- Complete blood count (CBC)

Additional testing for antithyroglobulin and antithyroid peroxidase (TPO) antibodies should be considered, especially if thyroid dysfunction is suspected.

Antithyroid peroxidase antibodies are particularly sensitive and specific markers for autoimmune thyroid disease.

Histology

Histopathologically, vitiligo is defined by the absence of melanocytes in lesional epidermis. At the margins of active lesions or in early lesions, a superficial dermal perivascular and perifollicular lymphocytic infiltrate may be observed, supporting a cell-mediated immune mechanism targeting melanocytes.

Prognosis and Clinical Course

The prognosis and progression of vitiligo are unpredictable. The initial clinical subtype does not reliably predict the pattern of future skin involvement or disease activity.

Treatment

Multiple treatment options are available for vitiligo (see Box 72-2). Most aim to restore skin pigmentation. Each therapeutic approach has distinct advantages and limitations, and no single treatment is suitable for all patients.